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Div: D

Subject: Full stack Development Lab

Assignment 6

Assignment : Online Portal Development using Node.js, Express, and NoSQL

A. Node.js: The Runtime Environment

Node.js is an open-source, cross-platform runtime built on the Chrome V8 engine. Its core advantage is the **Single-Threaded Event Loop** and **Non-blocking I/O**, which allows the server to handle thousands of concurrent connections without the overhead of thread management. In an e-commerce or booking context, this ensures that the server remains responsive even during high-traffic events like flash sales or peak booking hours.

B. Express.js: The Backend Framework

Express is a "unopinionated" web framework that provides a robust set of tools for building web applications.

- **Routing:** It maps incoming HTTP requests (GET, POST, etc.) to specific controller logic.
- **Middleware:** It allows for modular processing of requests, such as verifying user credentials or parsing incoming JSON data before it reaches the final application logic.

C. NoSQL and MongoDB: Flexible Data Storage

Unlike traditional relational databases, **MongoDB** is a document-oriented NoSQL database that stores data in **BSON (Binary JSON)** format.

- **Schema Flexibility:** Used items (cars, bikes) often have varying attributes; NoSQL allows for dynamic fields without requiring a rigid table structure.
- **High Availability:** MongoDB supports replica sets, ensuring that the application remains functional even if a database node fails.
- **Mongoose ODM:** Mongoose is used to provide a schema-based solution for application data, handling validation and casting directly within the Node.js environment.

3. Application Architecture

The application follows a **Three-Tier Architecture**:

1. **Presentation Layer (Frontend):** Developed with HTML, CSS, and JavaScript, it captures user input and displays data fetched from the server.

2. **Logic Layer (Backend):** The Express server processes business logic, such as calculating prices or checking appointment slots.
3. **Data Layer (Database):** MongoDB persists user accounts, item listings, and transaction history.

4. Implementation Components

I. RESTful API Endpoints

Communication between the frontend and backend is facilitated through REST APIs:

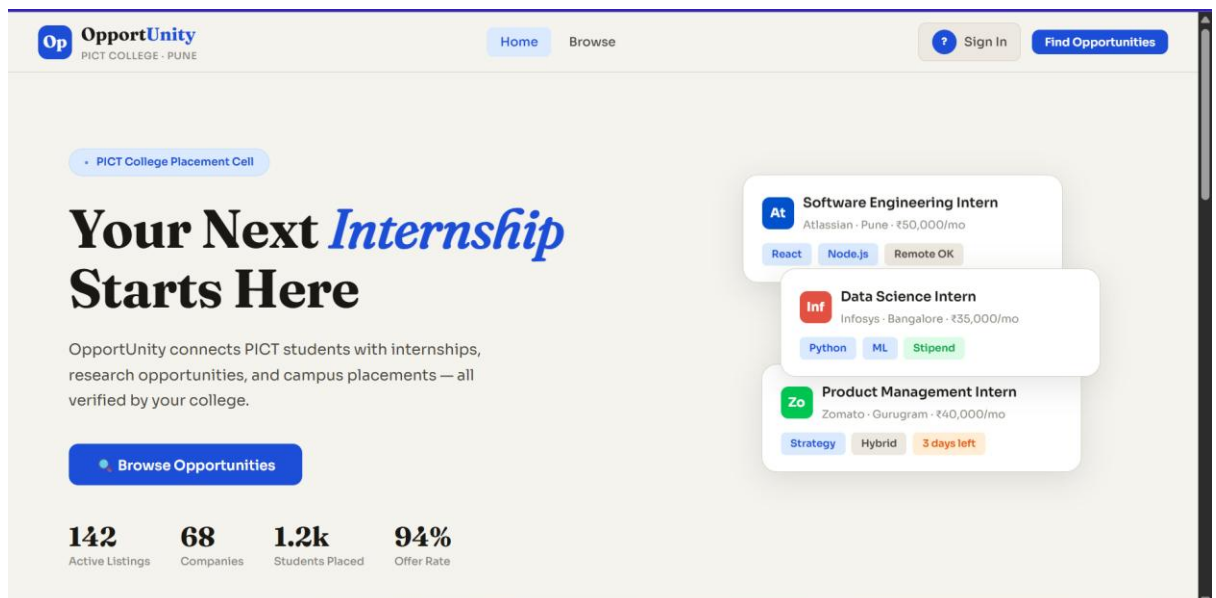
- **GET /api/items:** To browse available cars, bikes, or appointment slots.
- **POST /api/bookings:** To submit a new purchase request or book an appointment.
- **PUT /api/items/:id:** To update item details or mark a slot as "unavailable".

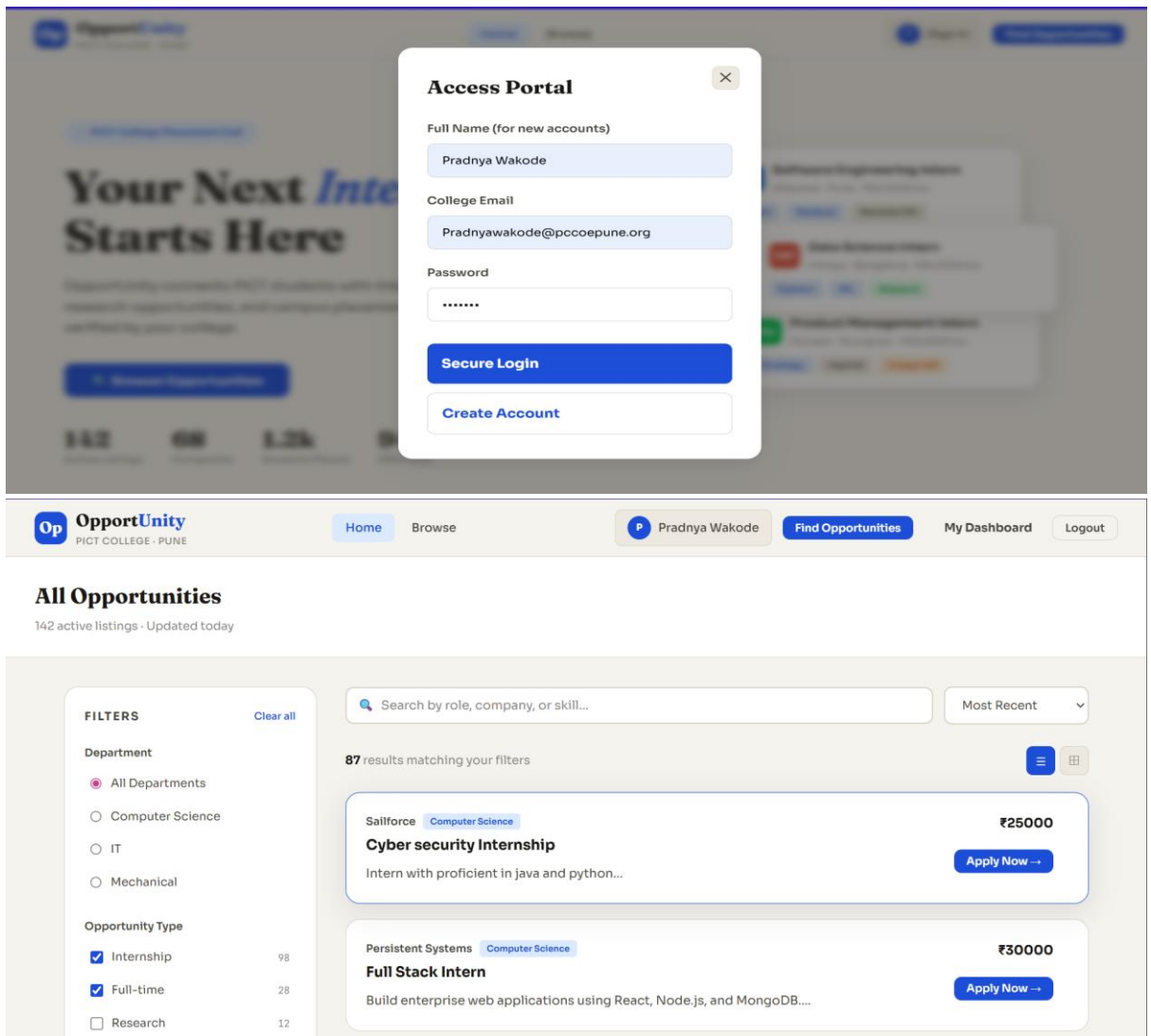
II. Authentication and Security

- **Password Hashing:** The system uses bcrypt to securely hash passwords before they are stored in the database.
- **JWT (JSON Web Tokens):** Upon a successful login, the server issues a JWT. This token is sent with every subsequent request to identify the user and their permissions (e.g., student, customer, or admin).

III. File and Data Handling

- **Multer Middleware:** For e-commerce portals, Multer handles the upload of item images (e.g., photos of used cars) or PDF documents for verification.
- **Static Assets:** Express is configured to serve these uploaded files and other static assets from a public directory.





Results:

Admin Portal:

Access Portal

Full Name (for new accounts)

Admin

College Email

Internshipadmin@pccoepune.org

Password

.....

Secure Login

Create Account

Op

OpportUnity
PICT COLLEGE - PUNE

Home

Browse

Post Internship

Admin

Find Opportunities

Admin Dashboard

Logout

Central Placement Admin

Post New Opportunity

Refresh Applications

STUDENT	INTERNSHIP	ACTION/STATUS		DOCUMENT
Pranja; pranjal@pccoepune.org	Cyber security Internship Sailforce - Computer Science	Pending	Resume PDF	
Pradnya Wakode Pradnyawakode@pccoepune.org	Cyber security Internship Sailforce - Computer Science	Shortlisted	Resume PDF	
Pradnya Wakode Pradnyawakode@pccoepune.org	Full Stack Intern Persistent Systems - Computer Science	Shortlisted	Resume PDF	
Pradnya Wakode Pradnyawakode@pccoepune.org	Frontend Engineering Intern Google - All	Shortlisted	Resume PDF	
Pradnva Wakode	Data Science Intern			

Company Name

Bajaj Fiserv

Internship Title

Full Stack Intern

Engineering Department

IT

Location

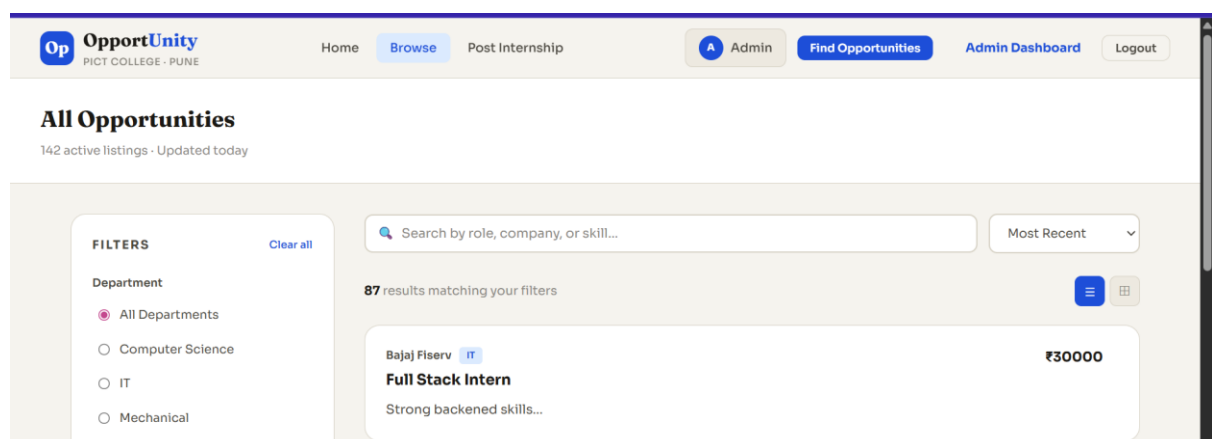
Pune

Stipend (Monthly)

30000

Role Description

Strong backened skills



5. Conclusion

Integrating Node.js, Express, and MongoDB creates a versatile platform capable of handling diverse use cases from e-commerce to service booking. The combination of an asynchronous backend and a flexible NoSQL database provides the agility required to adapt to changing user requirements and data structures in a modern digital marketplace.